

End Semester Exam: Advanced Optics Lab. PHY649 21-11-2025

• **Attempt ANY FOUR questions. Each question carries 6 marks. Duration: 1 Hr**

1. A nanosecond green laser emits 20 ns pulses at 1.0 kHz rep rate and 1 mW average power. Calculate the following: (i) peak power in the laser beam, (ii) Its spectral width, (iii) length of single pulse in space, and (iv) distance between two consecutive pulses in space.
2. Explain key components of a green CW diode pumped solid state (DPSS) laser. How many different types of photons are there in the cavity? What are special transmission/reflection properties of the cavity mirrors to operate this laser? Explain with a diagram.
3. Design an experiment to measure coefficient of absorption of a solid medium? Mention its units. The red coloured acrylic is exposed to (i) 20 ns pulses@ 1kHz rep rate with 1mW average power. In another experiment, the same medium is exposed to a cw laser of 1mW peak power. Keeping all other parameters identical which laser (pulsed or CW) will create more damage to solid material and why.
4. Draw a labelled schematic diagram of the apparatus to measure efficiency of a solid-state laser? Draw a typical V-I curve of a laser.
Explain the important difference in spectral properties of laser beam just below and just above the lasing threshold. You may ignore the power difference.
5. Mention six key parameters of a laser beam. Name the instrument/technique to measure these parameters.
6. Describe the working principal of a semiconductor red/blue diode laser. Draw a band diagram to explain how the population inversion takes place.
Design an experimental setup to measure efficiency of the diode laser? Label each component used in the setup.